IE UNIVERSITY | SCHOOL OF SCIENCE & TECHNOLOGY

Machine Learning & Communication Skills – Sentence Outline

FIELD: How Much Does the Field of Study Shape University GPA Rankings?

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FIELD: Field Influence on Evaluation Levels across Degrees.

1 Introduction

1. [Adam: universities often allocate scarce places by ranking students from all degrees on one GPA list, which assumes a grade means the same in every degree.]
   a. [Adam: grading standards differ between departments of the same university [1, 2].]
   b. [Adam: the gaps are large and repeat across 33 US public universities [3].]
2. [Adam: research question, stated as a claim: we measure how much of a student's position in a pooled GPA ranking depends on the field of study, and how ranking within degree changes who is selected.]
   a. [Adam: what the project measures (grading differences between fields) and what it does not claim (intrinsic difficulty of a degree).]
   b. [Adam: one sentence that previews the main finding and the structure of the report.]

2 Data

3. [Pablo: three complementary sources answer different parts of the question, and they are analysed separately because they share no key.]
   a. [Pablo: Spanish SIIU table: 12,125 university-field-year means, 89 universities, 29 fields, 2015-16 to 2023-24, with quartiles [4]; limits: aggregated, graduates only.]
   b. [Pablo: Portuguese student records: 3,703 students in 15 degrees with standardised entry grades [5].]
   c. [Adam: US data: what the 33-university dataset adds and how departments were mapped to fields [3].]

3 Methodology

4. [Pablo: the analysis separates shared field effects from university-specific ones and tests them on data the models have not seen.]
   a. [Pablo: variance decomposition with fixed effects, re-estimated with a mixed-effects model with random university and university-by-field effects.]
   b. [Pablo: temporal validation (train 2015–2021, test 2022–2023) with bootstrap intervals over universities.]
   c. [Pablo: confounders (standardised admission grade, size, dropout) and why dropout is treated separately.]
5. [Pablo/Nacho: the selection simulation and the comparison of ranking rules translate grade gaps into places won or lost.]
   a. [Pablo: the simulation approximates within-field grades from mean and quartiles, and its normal assumption was checked against reported medians and maxima.]

b. [Nacho: four ranking rules are compared by ranking on first-semester grades and evaluating them on second-semester grades.]

4 Results and Discussion

6. [Nacho: most of the variation in grades within a university is explained by the field of study, and part of it is specific to each university.]
   a. [Nacho: 65% shared field effect, 28% university-specific, 7% noise; the mixed model gives 61 / 29 / 11%.]
   b. [Nacho: the university-specific part is stable over time (r = 0.84, 95% CI 0.84–0.85).]
   c. [Nacho: past field profiles predict later years best (test RMSE 0.137 vs 0.226 for the shared effect alone).]
7. [Nacho: intake differences do not explain the field gaps.]
   a. [Nacho: admission grade and size leave field effects almost unchanged (SD 0.31, correlation 0.99); in the Portuguese records the gap only falls from 1.51 to 1.30.]
   b. [Nacho: dropout explains part of the spread, but low grades may themselves cause dropout.]
8. [Adam: the US replication result: whether the ranking of fields is similar in both countries (correlation of field effects).]
   a. [Adam: main supporting number.]
   b. [Adam: main difference or limitation of the comparison.]
9. [Nacho: a single raw-GPA ranking gives very unequal access across fields, and the result is robust to the distributional assumption.]
   a. [Nacho: 41% of Nursing graduates vs 6% in Business Administration reach a top-20% cut (median university-year, with interquartile ranges).]
   b. [Pablo: replacing the normal with a truncated normal changes shares by 0.04 percentage points on average and no field changes rank.]
10. [Nacho: within-degree rules equalise access at almost no cost in later performance.]
    a. [Nacho: second-semester grade of the selected students 14.21 with raw GPA vs 14.08 with percentiles, overlapping 95% intervals.]
    b. [Nacho: spread of selection rates across degrees falls from 0.090 to 0.006.]
    c. [Nacho: the higher stability of raw GPA partly reflects the persistent field gap itself.]
11. [Nacho: the main limitations and what they mean for the conclusions (aggregated data, graduates only, public universities for admission grades).]

5 Conclusions and Future Work

12. [Nacho: synthesis: pooled GPA rankings sort students largely by field, and within-degree normalisation reduces this at little cost.]
13. [Nacho: future work, e.g. individual-level data from more institutions to separate grading practice from student ability.]

References

[1] R. Sabot and J. Wakeman-Linn, "Grade inflation and course choice," *Journal of Economic Perspectives*, vol. 5, no. 1, pp. 159–170, 1991, doi:10.1257/jep.5.1.159.

[2] A. C. Achen and P. N. Courant, "What are grades made of?," *Journal of Economic Perspectives*, vol. 23, no. 3, pp. 77–92, 2009, doi:10.1257/jep.23.3.77.

[3] H. Jeong, G. M. Weiss, A. Leung and D. D. Leeds, "The construction and analysis of course grades across public universities," in *Proc. 17th Int. Conf. Educational Data Mining*, 2024, pp. 643–648, doi:10.5281/zenodo.12729906.

[4] Ministerio de Ciencia, Innovación y Universidades, "Media, cuartiles y máximo de la nota del expediente (estudiantes egresados) por ámbito de estudio y sexo," SIIU, 2025. [Online]. Available: https://estadisticas.ciencia.gob.es

[5] V. Realinho, M. Vieira Martins, J. Machado and L. Baptista, "Predict students' dropout and academic success," UCI Machine Learning Repository, 2021, doi:10.24432/C5MC89.

Declaration of Contributions and AI Use

[All: one entry per member describing what each actually did (analysis, code, sentences, reviews), and every use of AI tools with tool, purpose, output used and verification.]